MIL-A-48073 (MU)
2 June 1972
SUPERSEDING
PA-PD-2798A
10 March 1969

MILITARY SPECIFICATION

ACCESSORIES, LOADING, FOR 152MM, CARTRIDGES M409 SERIES, M411 SERIES, XM617, M625 SERIES, AND XM657

1. SCOPE

1.1 This specification covers accessory parts required for loading of Cartridges M409 Series, XM411 Series, XM617, XM625 Series and XM657 for 152MM Gun (See 6.5).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-I-45607	-	Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-1167	-	Ammunition Data Card
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1169	-	Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1320

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FEDERAL

FED-STD-151 - Metals; Test Methods

DRAWINGS

ARMY

9204038 - Primer, Electric, M91
9204177 - Plug, Closing
9204182 - Tube
9204197 - Projectile, HEAT-MP-T, 152MM, M409
9211868 - Washer, Felt
9211869 - Pad, Shock
9223763 - Projectile, HE-T, 152MM, XM657E2
9223768 - Pad, Shock
9225370 - Cap, Terminal
9233521 - Strip
9233541 - Primer, Electric, XM117
9234190 - Tube
9244308 - Ring, Case Mounting

PUBLICATIONS

ARMY

ET-9204177 - Equipment Tabulation
ET-9204182 - Equipment Tabulation
ET-9204197 - Equipment Tabulation
ET-9211866 - Equipment Tabulation
ET-9211867 - Equipment Tabulation
ET-9211868 - Equipment Tabulation
ET-9211869 - Equipment Tabulation
ET-9223768 - Equipment Tabulation
ET-9225370 - Equipment Tabulation
ET-9233521 - Equipment Tabulation
ET-9234190 - Equipment Tabulation
ET-9244308 - Equipment Tabulation

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Parts.-Parts shall comply with all requirements specified on the drawings specified in 2.1 as applicable, and all the requirements specified in applicable specifications.

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3.3 Workmanship.-All parts shall be dry and free of chips, dirt, grease and other foreign material. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

3.4 First Article Inspection.-This specification makes provisions for first article inspection. Requirements for submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of Product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

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The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 First Article Inspection

4.1.2.1 Case Mounting Ring

4.1.2.1.1 Submission.-The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.1.2.1.2. The first article sample shall consist of 50 case mounting rings and two tensile test specimens (See 4.2.3.1.2) which have been produced by the contractor or furnished by a supplier, and which have been manufactured using the same production process, procedures and equipment which will be used in fulfilling the contract. All parts and materials shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract, and submit a record of this inspection with the sample, including statements of findings for materials, processes and tests. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, materials used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.1.2.1.2 Inspection to be Performed.-The sample will be subjected, by the Government, to any or all of the examinations and any or all requirements of the applicable drawing.

4.1.2.1.3 Rejection.-If any part or assembly fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component to comply with any of the stated requirements.

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4.2 Quality Conformance Inspection

4.2.1 Lot Formation.-The term "inspection lot", as used in this specification, is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product as specified in Standard MIL-STD-105. Changes to the process, specifications or drawings, not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.2.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for Critical Defects (and Major Defects, when so specified) shall be 100 percent. Sampling plans for Major and Minor Defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of Major and Minor Defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each Major Defect and an AQL of 0.65 percent for each Minor Defect. When this method is elected, it shall be applied to all of the Major and Minor Defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

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4.2.2.1 Cap, Terminal (See Drawing (dwg.) 9225370, covering a detail of dwg. 9204197).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 2.5 percent		
201.	Outside diameter	Gage	01001
202.	Thickness of material at back of cavity	Gage	01002
203.	Width, total	Gage	01003
204.	Presence of slot	Visual	01004
205.	Evidence of poor workmanship (See 3.3)	Visual	01005

4.2.2.2 Tube (See dwg. 9204182, covering a detail of dwg. 9204038 or dwg. 9234190, covering a detail of dwg. 9233541, as applicable).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Length, maximum (max.)	Gage	02001
102.	Inside diameter	Gage	02002
103.	Scrim exposed on inner or outer surface	Visual	02003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship	Visual	02004

4.2.2.3 Plug, Closing (See dwg. 9204177, covering a detail of dwg. 9204038).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		

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Major:	AQL 0.65 percent		
101.	Large outside diameter.....	Gage	03001
102.	Small outside diameter.....	Gage	03002
103.	Length.....	Gage	03003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship.....	Visual	03004

4.2.2.4 Washer, Felt (See dwg. 9211868, covering a detail of dwg. 9204197).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major: None defined.			
Minor:	AQL 1.5 percent		
201.	Outside diameter, max.	Gage	04001
202.	Inside diameter, minimum (min.).....	Gage	04002
203.	Evidence of poor workmanship.....	Visual	04003

4.2.2.5 Pad, Shock (See dwg. 9211869, covering a detail of dwg. 9204197, or dwg. 9223768, covering a detail of dwg. 9223763, as applicable).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Color incorrect.....	Visual	05001
Minor:	AQL 1.0 percent		
201.	Outside diameter, max.	Gage	05002
202.	Evidence of poor workmanship.....	Visual	05003

4.2.2.6 Strip (See dwg. 9233521, covering a detail of dwg. 9204197).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			

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Major:	AQL 0.65 percent		
101.	Length	Gage	06001
102.	Thickness	Gage	06002
103.	Width	Gage	06003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship	Visual	06004

4.2.2.7 Ring, Case Mounting (See dwg. 9244308).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.0 percent		
101.	Location of datum diameter	Gage	07001
102.	Thickness of ring	Gage	07002
103.	Inside diameter	Gage	07003
104.	True position of inside diameter to the datum diameter	Gage	07004
105.	Presence of burrs	Visual	07005
106.	Angle tangent to radius	Gage	07006
Minor:	AQL 2.5 percent		
201.	Height of ring	Gage	07007
202.	Inside radius	Gage	07008
203.	Improper cleaning prior to application of protective finish	Visual	07009
204.	Bare metal exposed after application of protective finish	Visual	07010
205.	Evidence of poor workmanship	Visual	07011

4.2.3 Testing

4.2.3.1 Ring, Case Mounting

4.2.3.1.1 Hardness (See dwg. 9244308) - Major Defect - Code No. 07012 - AQL 0.40 percent.-Samples shall be randomly selected from each heat treatment lot (See 6.4) in accordance with MIL-STD-105, General Inspection Level II and subjected to a hardness test as specified in 4.3.1.1. Failure to comply with the applicable requirements shall reject the heat treatment lot which the samples represent.

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4.2.3.1.2 Tension - For First Article Information Only.-Two (2) tensile test samples shall be taken from the same coil of strip steel as the case mounting rings in the first article and shall be heat treated along with the first article. These specimens shall be subjected to the yield strength test in accordance with 4.3.1.2.

4.2.4. Inspection Equipment.-Equipment Tabulation Numbers, as listed in paragraph 2.1, identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 Government Rights to Documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.2.4.2 Supply and Maintenance.-Supply and maintenance of the equipment listed on the ET shall be in accordance with Specification MIL-I-45607.

4.2.4.3 Government Use of Contractor's Inspection and Test Equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test Methods and Procedures

4.3.1 Ring, Case Mounting

4.3.1.1 Hardness.-The test shall be performed after final heat treatment with a Rockwell Hardness tester or equal. The test shall be performed in a manner prescribed in Federal Test Method Standard No. 151, Method 243.

4.3.1.2 Yield Strength.-The largest obtainable standard specimen, as described by Federal Test Method Standard No. 151, Method 211, shall be taken from the tensile test samples described in 4.2.3.1.2. The criteria for determining yield strength shall be as described in Federal Test Method Standard No. 151, Method 211.

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5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level C.-All parts shall be packed in such a manner as to afford protection against damage during shipment from the supply source to the first receiving activity and in accordance with Standard MIL-STD-1169.

5.2 Marking.-In addition to any special marking which may be required by the contract or order, marking shall be in accordance with Standard MIL-STD-1169.

6. NOTES

6.1 Ordering Data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data Cards - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Provisions for submission of first article samples.

6.2 Inspection Code Numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are used to correlate the characteristics cited on the Equipment Lists with the inspections listed in this specification. They should also be cited as references on drawings of equipment designs submitted by the contractor (See 6.3.5).

6.3 Inspection Equipment.-The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection Equipment Lists (EL).-Inspection Equipment Lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes, as indicated in paragraph

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6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection Equipment List Codes.-The inspection equipment, as defined in 6.3.3 and 6.3.4, will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.

MU - Army design, Mandatory for Use

6.3.3 Army Designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor-prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor Designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which completely depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance

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with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category A, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of Contractor Designs.-All submitted designs shall contain a reference to the applicable five-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as Critical and Major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-A. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged; however, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Heat Treatment Lots

6.4.1 Case Mounting Ring Heat Treatment Lot (Batch Process).-The heat treatment lot shall consist of case mounting rings manufactured from one lot of strip steel which has been heat treated in the same set of furnaces, at the same time, as a group. After final heat treatment, case mounting rings from the same heat treat lot shall remain grouped together for identifying purposes until the results of the physical and mechanical properties tests prescribed indicate compliance with the applicable specification and drawing requirements.

6.4.2 (Alternate) Case Mounting Ring Heat Treatment Lot (Continuous Process).-The heat treatment lot shall consist of case mounting rings manufactured from one lot of strip steel which has been heat treated in the same set of furnaces during no more than a continuous 16-hour period under controlled conditions. If during this time period the furnace is shut down or conditions change, the case mounting rings heat treated prior to the occurrence shall be considered the heat treatment lot.

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After final heat treatment, case mounting rings from the same heat treatment lot shall remain grouped together for identifying purposes until results of the physical and mechanical properties tests prescribed indicate compliance with the applicable specification and drawing requirements.

6.5 Usage of Components

ACCESSORY PARTS

		CARTRIDGE				
Part Number	Part Name	M409 Series	M411 Series	XM617	M625 Series	XM657
9204177	Plug, Closing	x	x	x	x	x
9204182	Tube	x	x	x	x	
9211868	Washer, Felt	x				
9211869	Pad, Shock	x				
9223768	Pad, Shock					x
9225370	Cap, Terminal	x				
9233521	Strip	x				
9234190	Tube					x
9244308	Ring, Case Mounting	x	x	x	x	

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1320-A138

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